27 Matrix Find a specific pair in matrix

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#include <bits/stdc++.h>
using namespace std;
class Solution {
  public:
    int findMaxValue(vector<vector<int>>& mat, int N) {
        if (N <= 0) return 0;
        // maxArr[i][j] will store maximum element in submatrix (i..N-1, j..N-1)
        vector<vector<int>> maxArr(N, vector<int>(N, INT_MIN));
        // start from bottom-right
        maxArr[N-1][N-1] = mat[N-1][N-1];
        // last row
        for (int j = N-2; j >= 0; --j) {
            maxArr[N-1][j] = max(mat[N-1][j], maxArr[N-1][j+1])   }
        // last column
        for (int i = N-2; i >= 0; --i) {
            maxArr[i][N-1] = max(mat[i][N-1], maxArr[i+1][N-1]);
        int maxDiff = INT_MIN;
        // fill rest and compute max difference using maxArr[i+1][j+1]
        for (int i = N-2; i >= 0; --i) {
            for (int j = N-2; j >= 0; --j) {
                // candidate difference using bottom-right region
                int candidate = maxArr[i+1][j+1] - mat[i][j];
                if (candidate > maxDiff) maxDiff = candidate;
                // compute maxArr[i][j]
                int curMax = mat[i][j];
                curMax = max(curMax, maxArr[i][j+1]);   // right
                curMax = max(curMax, maxArr[i+1][j]);   // down
                curMax = max(curMax, maxArr[i+1][j+1]); // diag
                maxArr[i][j] = curMax;
             if no valid pair found (N==1) maxDiff remains INT_MIN; problem expects something sensible
        if (maxDiff == INT_MIN) return 0;
        return maxDiff;
    }
};
```